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21 September 1982
SUPERSEDING
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MILITARY SPECIFICATION

BOOTS, COMBAT, LEATHER, BLACK, (MIL-5)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers the requirements for a black full-laced leather combat boot.

1.2 Classification. The boot shall be of one type in the following sizes and widths as specified (see 6.2).

TABLE I. Schedule of sizes

| Widths | 3 | 3 1/2 | 4 | 4 1/2 | 5 | 5 1/2 | 6 | 6 1/2 | 7 | 7 1/2 | 8 | 8 1/2 | 9 | 9 1/2 | 10 | 10 1/2 | 11 | 11 1/2 | 12 | 12 1/2 | 13 | 13 1/2 | 14 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| XN | - | - | - | - | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| N | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| R | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| W | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |
| XW | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X | X |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Natick Research and Development Laboratories, Natick, MA 01760, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | | |
|---|---|---|
| V-L-61 | - | Laces, Nylon |
| V-T-285 | - | Thread, Polyester |
| V-T-295 | - | Thread, Nylon |
| KK-I-570 | - | Insole, Footwear, Leather, Cattlehide |
| KK-L-165 | - | Leather, Cattlehide, Vegetable Tanned and Chrome Retanned, Impregnated: and Soles |
| KK-W-231 | - | Welting, Leather, Shoe |
| TT-C-490 | - | Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings |
| UU-P-268 | - | Paper, Kraft, Wrapping |
| ZZ-H-141 | - | Heels, Rubber |
| CCC-C-443 | - | Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat) |
| DDD-T-86 | - | Tape, Textile; Cotton, General Purpose (Unbleached, Bleached, or Dyed) |
| PPP-B-636 | - | Boxes, Shipping, Fiberboard |

MILITARY

| | | |
|---|---|---|
| MIL-L-3122 | - | Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored |
| MIL-L-10867 | - | Leather, Cattlehide, Gusset, Chrome Tanned, Fatliquored |
| MIL-C-13924 | - | Coating, Oxide, Black, for Ferrous Metals |
| MIL-S-40043 | - | Soles, Rubber, Shoe |
| MIL-C-41814 | - | Counter, Footwear |
| MIL-L-41815 | - | Lasts, Footwear, Men's Combat, US MIL-5 |
| MIL-C-43956 | - | Cloth, Twill, Cotton, 10.0 ounces (339 g) |

STANDARDS

FEDERAL

| | | |
|---|---|---|
| FED-STD-191 | - | Textile Test Methods |
| FED-STD-311 | - | Leather, Methods of Sampling and Testing |
| FED-STD-751 | - | Stitches, Seams, and Stitchings |



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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this document to the extent specified herein.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 36 - Brass Plate, Sheet, Strip, and Rolled Bar
B 134 - Brass Wire
D 5 - Penetration of Bituminous Materials
E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials
E 28 - Softening Point by Ring-and-Ball Apparatus

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 Material (see 6.4).

3.2.1 Leather. All leather components shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide.



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3.2.1.1 Upper leather vamps, quarters, and one-piece counterpocket-backstays. The leather for the vamps, quarters, and one-piece counterpocket-backstays shall be smooth plated, full grain, or corrected grain side leather conforming to treatment C of MIL-L-3122, except that the thickness shall be as specified for the cut parts in 3.6.1 and the counterpocket-backstay may be split or shaved to the specified thickness. The break for vamps and quarters shall not be more than a break pattern of 4 when examined as specified in 4.2.3.1. The color shall be black. All finishes applied by the boot manufacturer and tanner shall be compatible with one another to insure a long-lasting lustrous finish.

3.2.1.2 Leather tongues, eyelet stays, and top facings. Tongues, eyelet stays, and top facings shall be gusset leather conforming to treatment C of MIL-L-10867 except that the thickness shall be as specified in 3.6.1 and that thinner pliable portions of leather, conforming to MIL-L-3122 and split to a thickness of not less than 2-1/4 ounces nor more than 3-1/4 ounces, may be used for eyelet stays and top facings. The color shall be black.

3.2.1.3 Counters. Counters shall conform to the requirements of MIL-C-41814 for boots, combat, leather, black, (MIL-5).

3.2.1.4 Midsoles, leather. The midsoles shall be full length, full grain leather conforming to types I, III, or IV, class 3, style 1, subclass b of KK-L-165. The midsole shall be not less than 6-1/2 irons or more than 8 irons in thickness after being leveled.

3.2.1.5 Welting. The welting shall conform to the requirements of KK-W-231.

* 3.2.1.6 Insole. The insole shall be cut from leather conforming to type I, tannage a or b, class 1 of KK-I-570. The insoles shall have either a channeled or stuck-on rib (see 3.2.4.4). The thickness of the channeled insoles shall be 6-1/2 irons and the thickness of the stuck-on rib insoles shall be 5-1/2 irons. As an alternate, the insole may be cut from Texorist material. The Texorist shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The insole shall be 0.116 $\pm$ 0.010 inch thick. The insole shall be cut with the heel-to-toe direction across the machine direction of the Texorist. The Texorist insole shall have a stuck-on rib applied to the printed side of the material (see 3.2.4.4).

* 3.2.1.7 Heel pads. The heel pads shall be cut from either chrome tanned or a combination of chrome and vegetable tanned, full grain or partially corrected lining leather, sheepskin, kip, calf, cattlehide, or cattlehide splits. The heel pads shall have a minimum thickness of 2 ounces and a maximum thickness of 3-1/2 ounces when tested as specified in 4.2.1. The color of the heel pads shall be either light russet or the color imparted by tanning process. Heel pads cut from upper leather or gusset leather specified in 3.2.1.1 and 3.2.1.2 may be used provided the top finish has been removed by buffing or splitting, and the thickness is a minimum of 2 ounces and a maximum of 3-1/2 ounces when tested as specified in 4.2.1. The heel



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pads shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.2.1. As an alternate to the above, the heel pads shall be chrome tanned or chrome vegetable retanned split pigskin. The pigskin shall be the first split from the grain side after the grain has been removed. The thickness shall be 2 to 3 ounces. The pigskin shall be drum dyed and the color shall be light russet. The surface of the split that was nearest the grain surface shall be sealed with a clear lacquer, sufficient to coat the surface fibers but not decrease porosity and then smooth plated. The sealed plated surface shall be worn next to the foot. The pigskin leather shall contain 0.18 to 0.70 percent paranitrophenol when tested as specified in 4.2.1.

3.2.2 Rubber compounds.

3.2.2.1 Outsoles, rubber. The outsoles shall be black rubber nonmarking full soles conforming to type III, class 1 of MIL-S-40043 except that for boot sizes 7 through 12, types I or III, class 1 soles may be used.

3.2.2.2 Heels, rubber. The heels shall conform to type II, class 3, style 2, color No. 1 of ZZ-H-141.

3.2.3 Metal.

3.2.3.1 Steel shanks (see 3.2.7.2).

3.2.3.2 Nails, heel attaching and sole reinforcing. Heel nails shall be brass or steel, cut or wire type and shall be of sufficient length to produce a clinch on the insole. The cut type brass nails shall be cut from material conforming to alloy No. 6 or 8 of ASTM B 36. The nails shall be commercial number 450 or 1336 with a 20 gage point. Gaging of points shall be done with an English standard wire gage in which 20 gage is 0.035 $\pm$ 0.005. The points of nails should enter the gage at least 1/32 inch. The wire nails shall be commercial type 13-1/2 gage with a flat clinching point and brass nails shall be cut from material conforming to alloy No. 7 of ASTM B 134. Sole reinforcing nails shall be steel or brass. The brass nails shall conform to alloy No. 6 or 8 of ASTM B 36. The nails shall be commercial type No. 19 or "A" head, and shall be of sufficient length to produce a "J" clinch on the insole.

3.2.3.3 Tacks and staples. Heel seat lasting tacks, welt butting tacks, and tacks or staples for attaching shanks shall be brass or steel, shall be of sufficient length to secure the parts through which they are driven, and shall leave the insoles smooth on the inside. Brass tacks shall be cut from material conforming to alloy No. 6 or No. 8 of ASTM B 36.



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3.2.3.4 Eyelets. The eyelets shall be made of aluminum, 0.0160 ± 0.0015 inch thick, with roll setting barrel and shall conform to the following finished requirements when tested as specified in 4.2.1.

Outside diameter of flange - 0.480 to 0.496 inch
Outside diameter of barrel - 0.295 to 0.299 inch
Diameter of hole before setting - 0.235 to 0.240 inch
Overall length - 0.224 to 0.236 inch

3.2.3.4.1 Finish of eyelets. After fabrication, the eyelets shall be anodized and top roller coated or tumble coated with not less than two coats of black enamel.

3.2.4 Fabrics.

3.2.4.1 Channeled insole reinforcement. The insole reinforcement shall be a cotton duck cloth conforming to type I of CCC-C-443.

3.2.4.2 Vamp lining. The vamp lining shall be cut from cotton twill conforming to class 2 of MIL-C-43956.

3.2.4.3 Backseam tape. The tape shall conform to type I, class 1, 1/2 or 17/32 inch width of DDD-T-86.

3.2.4.4 Stuck-on rib (when used, see 3.2.1.6). The finished rib shall be 7/32 to 8/32 inch high, not less than 5/8 inch wide measured from the inside vertical portion of the rib, and shall extend from heel breast line to heel breast line on the insole. The rib shall be made from coated fabric or a combination of coated fabric and fiberboard. The fabric shall conform to requirements specified in 3.2.4.4.1. The upstanding portion of the rib shall be one of the following: not less than three plies of coated fabric folded over on itself, two layers of coated fabric with a fiberboard filler not less than 0.050 inch thickness, or fiberboard and coated fabric.

3.2.4.4.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table II. The fabric shall be coated on one side with a rubber-base adhesive specified in 3.2.11, and bonded to the flesh side of the insole. The fabric or fiberboard shall cover 5/32 -2/32 +1/32 inch of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide for the required edge extension on the finished boot. As an alternate, the fabric bonded to the flesh side of the insole between the ribs, having a pinked edge is permitted.



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TABLE II. Ribbing fabric requirements

| Characteristic | Requirement, minimum |
|---|---|
| Weight | 8.0 ounces per sq. yd. |
| Yarns per unit length: | |
| Warp | 52 yarns per inch |
| Filling | 30 yarns per inch |
| Breaking strength: | |
| Warp | 105 pounds |
| Filling | 105 pounds |

3.2.4.4.2 Rib strength. The physical requirements for the rib strength shall conform to table III when tested as specified in 4.2.2.2.1.

TABLE III. Rib strength requirements

| Characteristic | Minimum 1/ | Average |
|---|---|---|
| Shear strength | 70 pounds | 75 pounds |
| Stitch strength | 20 pounds | 30 pounds |

1/ No single determination (sample unit) shall fall below the minimum value specified and the average of all specimens (lot average) shall be not less than the average specified.

3.2.5 Thread.

3.2.5.1 Thread, upper fitting. The thread for upper fitting shall be nylon and shall conform to types I, II or III of V-T-295 except that requirements for colorfastness shall not apply. The thread size shall be E for needle and bobbin with stitch type 301, and E for the needle and B or E for the looper with stitch type 401. The color of the thread shall be black.

3.2.5.2 Thread, sole stitching. The thread for sole stitching shall be polyester. The color of the running thread shall be black and the color of the shuttle thread shall be natural. The polyester thread shall be size 10, 3 ply for the running thread and size 10, 3 ply or size 10, 6 ply for the shuttle thread; and shall conform to type I, class 1, subclass C of V-T-285. The requirements for colorfastness need not apply.



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3.2.5.3 Thread, inseaming. The inseaming thread shall be polyester thread conforming to type I, class 1, subclass C, size 10, 3 or 6 ply of V-T-285. The color shall be natural.

3.2.6 Bottom filler. The bottom filler shall be either thermoplastic or cold process type.

3.2.6.1 Thermoplastic bottom filler. The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder in proportion of a minimum of 2-3/4 parts by volume of cork to each one part of binder. The cork granules shall be free from bark. The ground cork and the binder shall be thoroughly and evenly mixed. The binder shall be water insoluble. The binder shall have a softening point of at least 125°F and a maximum penetration of 85 millimeters with a 200-gram load for 60 seconds at 77°F when tested as specified in 4.2.1.

3.2.6.2 Cold process bottom filler. The cold process type shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When spread filler is dry and set, it shall consist of four parts cork to one part binder. The binder shall be water insoluble and shall have a softening point of at least 150°F when tested as specified in 4.2.1. The binder shall be tested after evaporation to a constant weight level. As an alternate, a cold process bottom filler applied by semiautomatic equipment or method may be used. The alternate filler shall consist by weight of a mixture of one part ground cork to five parts suitable binder. Upon loss of solvent, the alternate filler shall consist by weight of one part cork to 3-3/4 parts binder. The binder shall be water insoluble and have a softening point of at least 125°F when tested as specified in 4.2.1.

3.2.7 Shank. The shank shall be made of a shank board piece with a single reverse rib steel shank securely attached by four prongs or two rivets. The steel shank shall be 1/2 ± 1/16 inch from the front edge of the shank board piece. There shall be 7 sizes of shanks for the run of boot sizes specified in 3.2.7.3.

3.2.7.1 Shank board. The shank board shall be water resistant (shank board that has been hot waxed is considered as being water resistant). The finished thickness of the shank board shall be 1/8 ± 1/32 inch. The shank board shall be cut in accordance with Government furnished patterns specified in 3.4. The shank board pieces shall be molded to follow the bottom cavity of the boots, and both ends shall be skived with a graduated scarf 1-3/8 ± 1/16 inches wide. The shank board pieces shall be marked consecutively with a steel stamp starting with figure "1" on the smallest size up to and including "7" on the largest size.

3.2.7.2 Steel shank. The steel shank shall be constructed from 19 gage, 0.0418 ± 0.0030 inch cold rolled carbon steel with a hardness ranging from 47 to 54 Rockwell C scale when tested as specified in 4.2.1 and shaped to the manufacturer's standard number 4 bend conforming to the arch of the US MIL-5 last. The width of the steel shank shall be 5/8 ± 1/64 inch. The steel shank shall be made with a



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single rib. The overall thickness of the steel shank with rib shall be 0.080 to 0.125 inch. The steel shank shall have a zinc electroplated coating, dull or bright finish, or a zinc compound coating completely and uniformly applied to the base metal following a cleansing by any method of TT-C-490. Steel shanks with black oxide coating conforming to class 1 of MIL-C-13924 may be used in lieu of steel shanks with zinc coating. The rib shall taper off at a point within 3/8 to 7/8 inch from each end. The steel shank for size 4 shall be 4-1/2 ± 1/16 inches in length and the lengths for the other sizes shall grade up or down 1/4 ± 1/16 inch.

3.2.7.3 Shank fitting sizes. The shank fitting shall conform to the schedule as specified in table IV.

TABLE IV. Shank fitting schedule

| Shank size | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
|---|---|---|---|---|---|---|---|
| Boot widths and sizes | | | | | | | |
| XN - N | 3 - 6-1/2 | 7 - 8-1/2 | 9 - 10-1/2 | 11 - 12 | 12-1/2 - 13-1/2 | 14 | |
| R - W | 3 - 5-1/2 | 6 - 7-1/2 | 8 - 9 | 9-1/2 - 10-1/2 | 11 - 12 | 12-1/2 - 13-1/2 | 14 |
| XW | 3 - 3-1/2 | 4 - 5-1/2 | 6 - 7-1/2 | 8 - 9-1/2 | 10 - 11-1/2 | 12 - 13 | 13-1/2 - 14 |

3.2.8 Box toes. The box toes shall be cut from a material consisting of a woven base flannel or processed (needled) fabric, impregnated with a pyroxylin or polystyrene compound. The box toes may be either activated or a prepared type. The thickness of this material shall be 0.0875 ± 0.0075 inch. The box toes shall be of a shape and size to engage the insole around its toe radius after lasting, and shall be skived 9/16 ± 1/16 inch at the breast.

3.2.9 Pegs, heel seat fastening. The heel seat fastening pegs shall be glue sized paper base fiber, commercial type B or equal, 0.081 inch to 0.111 inch diameter regardless of type of equipment used. The pegs shall be of sufficient length to firmly secure the parts through which they are driven and leave the insole smooth on the inside.

3.2.10 Laces. The laces shall conform to type II, class 3 of V-L-61. The color of the laces shall be black. The length of each lace shall be a minimum of 60 inches.



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3.2.11 Adhesive. The adhesive used for bonding the various parts of boots shall be of the following types:

(a) Natural rubber latex.
(b) Synthetic rubber latex (including chloroprene).
(c) Natural rubber solvent cements.
(d) Synthetic resin cements.
(e) Synthetic rubber solvent cement.

3.2.12 Wax, inseam sewing and sole stitching. The wax shall be white or golden and shall be a permanently plasticized resin. The wax in the temperature range of normal machine use shall thoroughly wax the threads described in 3.2.5.2 and 3.2.5.3.

3.2.13 Ink, edge setting. The ink shall produce a bright hard finish and shall be uniform in appearance after setting with a hot iron. The color of the ink shall be black.

3.2.14 Repairers. Repairers shall be liquid spray, crayon or paste-type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.2.15 Renovators. Renovators used in lieu of or in addition to repairers shall match the color of the upper leather. Application shall be by sponge or spray method.

3.2.16 Fillers. Fillers, where used, shall be capable of providing a foundation for the application of the top finish and shall be applied by sponge or spray method.

3.2.17 Top finish. The top finish shall be applied by either the sponge or spray method, and shall be a material compatible with and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall provide luster.

3.3 Design. The design shall be for a full lace closure with "V" notch in the eyelet row edge of the quarters, grain-out leather upper, plain toe vamp, cloth lined vamp and box toe, and one-piece leather counterpocket-backstay. The boot shall have a leather midsole, full length rubber composition outsole, and a whole rubber heel with beveled breast. (See figure 1).

3.4 Patterns and dies. The height of the finished boot, measure upward on outside from bottom at breast of heel to top of boot, shall be 10-1/2 $\pm$ 1/4 inches on size 9 and shall graduate up or down between whole sizes approximately 1/8 inch as indicated by patterns. A standard set of paper patterns will be furnished by the Government as a basis from which the contractor's dies or patterns shall be made.



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The patterns and contractor's cutting dies and patterns shall consist of the component parts as specified in table V. The Government patterns shall not be altered in any way and shall be followed except for lasting allowance, which shall be determined by the contractor. The lasting allowance around the heel seat may include two "V" notches no greater than 7/16 inch in depth at the center back area of the counterpocket. The contractor shall furnish wood insole rounding patterns conforming to paper patterns loaned by the contracting activity. The one-piece counterpocket-backstay may be used with an inverted "V" removed in the center of the back of the counterpocket. The height of the cut for this inverted "V" shall be a maximum of 1-1/4 inches from the bottom of the lasting allowance on a size 3 and grading upward to a maximum height of 1-1/2 inches on a size 14. The point of this inverted "V" shall be in the center back of the counterpocket. All outer lines of this counterpocket-backstay must conform to the Government furnished patterns except for this inverted "V". This "V" shall be removed in a curved line to conform, after closing, to the back part of the US MIL-5 last.

TABLE V. Contractor's dies and patterns

| Component parts | |
|---|---|
| Vamp | Whole and half sizes, all widths |
| Quarter ("V" notch) | Whole and half sizes, all widths |
| Counterpocket-backstay (one piece) | Whole and half sizes, all widths |
| Tongue | Whole and half sizes, all widths |
| Top facing | Five sizes, one width |
| Eyelet stay | Three sizes, one width |
| Vamp lining (cloth) | Whole sizes, all widths |
| Insole | Whole and half sizes, all widths |
| Boxtoe | 8 sizes, all widths |
| Shank board | 7 sizes, widths not applicable |

3.5 Lasts and markers. The boots shall be made on the US MIL-5 last (MIL-L-41815). All necessary sizes of lasts will be loaned to the contractor by the Government. A marker set consisting of a one-piece counterpocket-backstay, vamp, and quarter, in whole and half sizes, all widths, will be loaned to the contractor by the Government.

3.6 Construction.

3.6.1 Cutting uppers. The uppers shall be cut from grain-out leather. The leather shall conform to the thickness specified in table VI. Except for eyelet stay, top facing, and one-piece counterpocket-backstay, none of these parts shall be split or shaved. No parts cut off stretch shall be accepted. Vamps shall be cut from the bend area of the side leather.



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TABLE VI. Thickness of upper leather

| Part | Thickness, ounces | |
|---|---|---|
| | Minimum | Maximum |
| Vamp | 5 | 6 |
| Quarter | 4-1/2 | 6 |
| Counterpocket-backstay (one piece) | 4 | 6 |
| Eyelet stay and top facing | 2-1/4 | 3-1/4 |
| Tongue | 2-3/4 | 3-1/2 |

3.6.1.1 Skiving. The skiving of upper leather parts shall conform to the requirements specified in table VII.



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TABLE VII. Skiving requirements

| Part | Side | Location | Scarf width inches, (mm) 6 oz. leather | 5 oz. leather | 4-1/2 oz. leather | 4 oz. leather | Scarf edge thickness, millimeters, ± 0.3 mm |
|---|---|---|---|---|---|---|---|
| Quarter | Flesh | Back edge | 3/16 ± 1/32 | - | 1/16 ± 1/64 | - | 1.5 |
| Quarter | Flesh | Lasting edge 1/ | 5/8 ± 1/8 | - | 13/64 ± 3/64 | - | 1.5 |
| Quarter | Flesh | Top edge and eyelet row edge | 3/16 ± 1/32 | - | 3/32 ± 1/64 | - | 1.3 |
| Quarter | Flesh | Vamp margin 1/ | 3/16 ± 1/32 | - | 1/16 ± 1/64 | - | 1.5 |
| Vamp | Flesh | Throat and wings | 3/16 ± 1/32 | 1/8 ± 1/32 | - | - | 1.2 |
| Vamp | Flesh | Toe 1/ | 3/4 ± 1/8 | 9/16 ± 3/32 | - | - | 1.5 |
| Counter-pocket | Flesh | Lasting edge 1/ 2/ | 5/8 ± 1/8 | - | - | 3/16 ± 1/64 | 1.5 |
| Counter-pocket backstay | Flesh | Stitch edge | 3/16 ± 1/32 | - | - | 1/16 ± 1/32 | 1.3 |

1/ Skiving of the lasting edge of the counterpocket, toe vamp, quarter, and quarter vamp margin is optional.

2/ When one-piece counterpocket-backstay is 4 to 5 ounces thick, the skiving of the counterpocket lasting edge and stitch edge of the one-piece counterpocket-backstay shall not apply.

NOTE: A dash indicates that the scarf width is the same as indicated in the preceding column.





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3.6.2 Vamp, crimping. The vamp may be crimped, if necessary, to insure tight lasting.

3.6.3 Marking, permanent identification. The inside of each boot shall be marked on the grain side with the correct size and width, the contractor's identification symbol, and the month and year (expressed numerically) of the date of contract. The marking shall be impressed into the grain side of the leather in such a manner as to be permanently visible without cutting through at any point, and shall be placed approximately in the center of the inside quarter and between the two rows of top facing stitching, except that sizes 3 and 3-1/2 on all widths the grain side of the tongue instead of the quarter may be marked not less than 3/4 inch and not more than 1 inch from the top center of the tongue. Figures shall be Arabic and letters Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height. The contractor's symbol shall be in a block as shown by the following example: 10 W /AB/ 12-81.

3.6.4 Upper leather fitting. Line marking patterns shall be used for all upper fitting. No die stab marking will be allowed. Quarters shall be closed at the back using stitch type 401 of FED-STD-751. The back seam shall be closed with stitching closely positioned to the edge of quarters on the grain side, and shall be rubbed down. The backseams shall be reinforced on the grain side with backseam tape, using stitch type 301. All other upper stitching shall be done using stitch type 301. All upper stitching shall be 8 to 10 stitches per inch.

3.6.4.1 One-piece counterpocket-backstay fitting. The one-piece counterpocket-backstay shall be stitched to quarters with two rows of stitching spaced 1/16 to 1/8 inch between rows on the side edges of the one-piece counterpocket-backstay to the top of the boot or across the top edge of the one-piece counterpocket-backstay. The counterpocket-backstay shall be reinforced with two additional inside rows of stitching around the counterpocket and extending to the backstay in conformance with the curvature of the backstay area as indicated by the marker patterns. The inside rows of stitching shall be spaced 1/16 to 1/8 inch between rows. The use of single or double needle machine is permitted. If inverted "V" seam counter is used, the "V" seam shall be closed, sewing upward from the lasting edge. The closing shall be done with one row of stitching using stitch type 401 and 8 to 10 stitches per inch. The "V" seam shall extend approximately 1/16 inch on the flesh side and shall be rubbed down. The butt side of the "V" seam shall be on the flesh side of the leather.

3.6.4.2 Tongue fitting. The quarter eyelet stay shall underlap the tongue and shall be lockstitched with two rows of stitching close to the edge of the tongue. The tongue shall be stitched to the vamp grain side out, with two rows of stitching, with one row of stitching not more than 3/32 inch from the edge of the tongue and a second row placed not more than 3/32 inch from the edge of the vamp, both of these rows to include stitching through the vamp lining. The tongue and eyelet facings shall be stitched to the quarters with edge and eyelet rows trimmed flush to 1/8 inch undertrim. The inside top facing, inserted flesh out, shall be fitted overlapping or under the eyelet stay and shall be undertrimmed and stitched with one row of stitching at the top edge. The bottom edge of the facing shall also be stitched to the quarters with one row of stitching.



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3.6.4.3 Eyeletting. On each quarter there shall be nine eyelets for all sizes. Four eyelets shall be spaced evenly from blucher nose to the bottom portion of the "V" notch of the quarter and the remaining five eyelets shall be spaced evenly from top portion of the "V" notch quarter to top of the quarter. The edges of eyelets shall be 1/8 to 5/16 inch from the edge of the quarter. The eyelets shall be securely and smoothly clinched.

3.6.4.3.1 Lacing for lasting. The machine lacing for lasting shall provide for a 1 ± 1/8 inch opening after lasting. The quarter shall be laced in the three lower pairs of eyelets. As an alternate, the quarters may be laced in the first and third or the two lower pairs of eyelets.

3.6.4.4 Vamping, barring. Vamping shall be done with two rows of stitching close on the edge of the quarter, plus two rows single or double machine stitched, spaced 5/32 to 1/4 inch between inside rows. The vamp lap shall extend beyond the second two space rows but not more than 3/16 inch. The bar row shall be uniformly spaced between, and parallel to, the two rows of vamping. The vamp bar shall consist of one or two rows and shall be 5/8 ± 1/16 inch in length on each blucher ear. As an alternate, the bar consisting of one row of stitching may be placed outside and above vamping, not more than 1/16 inch and parallel to the vamping row at each blucher ear and in the length specified above. The barring shall be done by single or double needle, or by the automatic method with tying or locking ends.

3.6.5 Sole leather stock fitting. The grain side of the leather midsoles and the reverse side of the rubber outsoles shall be roughed their full length with coarse abrasive paper. Two coats of chloroprene adhesive as specified in 3.2.11 shall be applied to the roughed grain side of the leather midsole. The first coat serves as a primer and shall be allowed to dry before application of the second coat. A single coat of the same adhesive shall be applied to the buffed mating surface of the rubber outsole. (The combined leather midsole and rubber outsole shall then be molded to conform to the correct contour of the bottom of the boot.) A boot which has been correctly inseamed, inseam trimmed, bottom filled, and made over the pertinent size and width of the MIL-5 last shall be used to check the contour of the molded sole combination. The contour shall be checked at the sole-laying operation, not less than 8 hours after sole molding. The combined leather midsole and rubber outsole shall have a minimum bond strength of 8 pounds per inch of width, when tested as specified in 4.2.2.2.2.

* 3.6.5.1 Leather insoles. Insoles shall be sorted for varying fibers to get uniform channeling and casing for even weight. The insoles shall be fleshed. The insoles shall be rounded to pattern. Die cut insoles will be accepted provided they conform to paper insole patterns loaned by the Government. A ± 1/64 inch tolerance from Government loaned insole patterns will be allowed.

* 3.6.5.2 Insole scoring. The lip scoring cut shall not penetrate to a depth exceeding 50 percent of the insole thickness and the scoring for channeled insoles or marking for stuck-on rib insoles shall be located 2-5/8 ± 1/16 inches from the heel end of a correctly rounded or die cut size 8R insole, and shall grade up and down from that size in accordance with the rate of grade obtained from a lip cutting and scoring machine.



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3.6.5.3 Insole channeling. The insole, except stuck-on rib insole, shall be channeled and may be chamfered. The channel leaf and outside lip shall be not less than one-third the thickness of the insole, and the channel leaf shall be slightly thicker than the lip. The in-between substance shall be 1/64 to 1/32 inch greater than the combined thickness of the leaf and lip. The channel margin shall be a minimum of 5/32 inch and the maximum shall not exceed the required edge extension on the finished boot (see 3.6.17). The lips shall be cemented with a latex adhesive specified in 3.2.11(a) or (b) and pressed together at lip setting to incline inward and be a minimum of 1/8 inch in height.

3.6.5.3.1 Assembly of stuck-on rib. When assembled, the stuck-on rib insoles shall meet the requirements of 3.2.4.4.

3.6.5.4 Insole backing. All insoles except stuck-on rib insoles (see 3.2.4.4.1) shall be reinforced by the application of cotton duck as specified in 3.2.4.1. The duck shall be coated on one side with an adhesive compound as specified in 3.2.11, and shall be bonded to the flesh side of the insole. The duck shall cover the entire area between the ribs from the score line forward, shall extend to the top of the upstanding rib, shall be firmly and smoothly fitted and bonded to the base and the lip, and shall be trimmed to the required rib height. The insole shall be cemented with any adhesive as specified in 3.2.11 on the flesh side prior to attaching the coated duck.

3.6.6 Lasting. Uppers may be conditioned by any suitable means except dipping in water. The correct size and width of uppers, lasts, and insoles shall be assembled. Insoles shall be tacked to the last with not less than five tacks: one of the five tacks shall be in the center of heel seat, one at shank, one at each side of the ball, and one at the toe. Edges of the insoles shall be flush with the last bottom at all points. Staples in lieu of tacks may be used for attaching the insole to the last. The heel seat of the insole shall be smooth and even with the heel seat of the last. Counters shall be assembled in accordance with sizing schedule so that they will fill the counterpockets, and the counters shall be such as to assure the inseam stitching catching the ends of the counter with one or more stitches. Counters shall be cemented on both sides. When leather counters are used, the flesh side of the leather counter shall be towards the inside of the boot. One-half $\pm$ 1/8 inch of the counter shall be turned over the last at the bottom. The correct size of pyroxylin or polystyrene type box toe specified in 3.2.8 shall be conditioned and inserted between the vamp lining and the vamp. A light coating of adhesive specified in 3.2.11 shall be applied between the vamp lining and the vamp at the inside shank of each boot. Uppers of the correct size and width shall be assembled to the lasts in a manner to provide for a wiped-in heel seat of approximately 1/2 inch. As an alternate, heel seat lasting may be done by any suitable method or equipment to assure a flat, secure heel seat. Uppers shall be drawn over the lasts with tension on the pulling-over machine to assure that the quarters at the blucher points and the vamps are down to the last, blucher noses are even, and box toes are aligned and mated for length. Sides of boots shall be spindled, drawing the upper snugly to the last, and then side-lasted so that when stapled, the upper will be held securely to the last. Side-lasting and inseaming



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operations may be performed simultaneously (see 3.6.7). The vamp lining shall be pulled smooth and tight without any tears. The heel seat shall be wiped-in and tacked and shall be flat and free from wrinkles. The toe shall be smoothly wiped-in and the toe wire or nylon filament attached securely around the toe at the base of the insole rib. As an alternate, any suitable equipment or method may be used to smoothly and securely attach the toe around the base of the insole rib. When the alternate methods are used for forepart and heel seat lasting, excess upper material shall be trimmed so as not to interfere with the cementing of the lasting operation.

3.6.6.1 Time allowance on lasts. The boots shall remain on the lasts a sufficient length of time to thoroughly dry the counters, uppers, and sole leather.

* 3.6.7 Inseaming. Inseaming shall be done by machine with the thread waxed and with proper tension on the thread, using a needle not larger than number 41, and with not less than 3-1/4 stitches to the inch. The welting shall be tempered to provide a tight seam. The welting shall be inseam stitched to the bottom of insole rib from rib end to rib end and shall catch in counter with one or more stitches on each end. As an alternate, the operation for side-lasting and inseaming may be performed simultaneously with any suitable equipment or method. Prior to heel seat lasting and inseaming each boot shall be spindled, forward edge of the counter pulled into position and the upper with its lining drafted snugly down to the last on both sides of the boot.

3.6.8 Tack and staple pulling, inseam trimming. All tacks and staples shall be removed from the bottom of insole and no broken tack or staple points shall remain. The inseam shall be closely trimmed from butt of welt to butt of welt without cutting or damaging stitches. The ends of welt shall be skived with 5/8 $\pm$ 1/8 inch bevel and tacked within the butt area. The welt shall be beaten out smoothly while in temper to lie flat around the entire forepart of the boot.

3.6.9 Shank fitting - bottom filling. The shanks shall be selected for size in accordance with shank fitting schedule specified in 3.2.7.3. The shanks shall be inserted in position with the shank board piece filling the cavity between the inside and outside ribs from the ball line rearward to the back of the heel seat. The shanks shall be attached with pitch, wax, or two brass tacks or staples, one at each side of the shank board piece back of the breastline. Shanks may be attached with steel tacks. The forward end of the shank shall be flush with the insole and fit the contour of the boot bottom rearward of the ball line. The bottom filler shall be firmly pressed into the insole channel from the toe to the forward end of the shank board piece with a uniform and even surface. The bottom shall present a flat smooth surface for sole laying. No bottom filler shall remain under the shank board piece or over the welt. Any excess cavities between the shank board and the inside rib and the heel seat area shall be filled with botom filler.

3.6.10 Sole laying. The boot bottoms shall be cemented from toe to heel breast line with any adhesive specified in 3.2.11. After the combined midsole and rubber outsole have been cemented and molded as specified in 3.6.5, the midsole shall be cemented over the entire surface and shall be laid on a sole-laying machine with pressure. The soles shall be of adequate size and shall be laid evenly to allow for the specified edge extension. The plaque (tread design) shall not be off-center on the finished boot by more than 1/4 inch.



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3.6.11 Rough rounding. Soles shall be uniformly rounded on a rough rounding machine for the specified edge extension of sole and welt as specified (see 3.6.17).

3.6.12 Sole stitching. Soles and welts shall be stitched together on a lock-stitch machine using waxed polyester thread size 10, 3 ply for the running thread and size 10, 3 ply or size 10, 6 ply for the shuttle thread with a No. 45 round blade chisel point awl and not larger than a No. 45 needle, and using 5-1/2 to 7-1/2 stitches per inch. The stitching shall be laid on the surface of the welt, close to the outer edge of the welt and the stitches shall extend under the heel on the finished boot. The stitching shall be with the lock (peg) just under the surface of the outsole. The edge gage shall not be removed too far ahead of the joint.

3.6.13 Heel seat fastening. The heel seat fastening shall be done by a fiber fastening machine using fiber pegs specified in 3.2.9. When fastening is accomplished using automatic (gang fastener) equipment, the pattern provided by the equipment will be acceptable. When the pegs are driven one at a time, there shall be not less than four to the inch. Pegs and nails shall be positioned at the edge of the insole and extend around the heel seat from welt butt to welt butt. Alternatively, the heel seat fastening may be done by using numbers 12, through 39 head, brass or steel loose nails or sole reinforcing nails specified in 3.6.19 and shall be of sufficient length to clinch on the insole. Nails, when used, shall be spaced not less than three to the inch and shall clinch properly on the insole.

3.6.14 Heel seat rounding. Heel seats, midsoles, and outsoles shall be smoothly rounded from welt butt to welt butt.

3.6.15 Leveling. Bottoms shall be progressively leveled while in temper on a leveling machine by means of metal rolls. The machine shall be adjusted to level the bottom of the sole from the toe end back to breast line. As an alternate, any suitable equipment may be used, that will adequately level the sole bottom while in temper.

3.6.16 Heeling. Heels, as specified in 3.2.2.2, shall be of correct size and shall be attached with 13 nails. The nails shall be of a length to insure a secure clinch on the insole. The heeling machine shall be equipped with angular die blocks MB 6466A, using drivers in the eight forward positions of the driver plate of a minimum length of 3-5/8 inches, and the five rear drivers in every case to be 1/16 inch longer than the forward drivers. This detail shall be carefully checked to insure that all nails are driven evenly against the metal washers in the heels, and clinch on the insole.

3.6.17 Edge trimming. Edges shall be trimmed using cutter No. 30x12-538K1-260 (12-tooth) or equivalent type to produce a round edge allowing for the required finished extension of outside ball to be not less than 3/16 inch, and the toe and inside ball to be not less than 1/8 inch, when measured at right angles from the upper.



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3.6.17.1 Edge setting. The leather portion of the edge, consisting of the welt and midsole, shall be stained with the ink specified in 3.2.13 to match the color of the upper leather and set on the edge setting machine and made smooth and bright. The top surface of the welts shall be black. No iron stain shall be permitted.

3.6.18 Heel finishing. The heels shall be trimmed and smoothly scoured. The leather heel seat shall be stained and finished to match the sole edge.

3.6.19 Sole reinforcing. The soles shall be reinforced with brass or steel No. 19 or A head nails just below the inside and outside ball line and around the toe just inside the seam in accordance with figure 2, with nails spaced 5/8 ± 1/8 inch between centers. The number of nails used for the sole reinforcing shall conform to table VIII (see figure 2). The nailing machine shall have a No. 2203-10 button to assure a 1/16 inch minimum, firm, smooth turned (J) clinch. The machine driver shall be set so that the nail heads will sink slighly below the surface of the outsole. No nails shall be driven thrcugh the inseam.

TABLE VIII. Sole reinforcing nails

| Boot sizes | Number of nails | | |
|---|---|---|---|
| | Toe | Inside shank | Outside shank |
| 3-4-1/2 | 4 | 4 | 3 |
| 5-7-1/2 | 5 | 4 | 3 |
| 8-12-1/2 | 5 | 5 | 4 |
| 13-14 | 6 | 6 | 5 |

3.6.20 Finishing.

3.6.20.1 Preparation. The boots shall be cleaned by removing accumulated dirt such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.6.20.2 Treeing. The boots shall be treed while on the lasts or on a treeing machine, using right and left tree feet conforming to US MIL-5 lasts. All wrinkles shall be removed. No material shall be used that may injure the leather or thread.

3.6.20.3 Final finish. Boots shall be repaired and filled. All raw edges, including vamp line, counterpocket-backstay, top, and front edges shall be stained to match the upper leather. Materials and methods shall be as specified in 3.2.13 to 3.2.17.

3.6.21 Nails, tacks, or staples. Nails, tacks, and staples that have been left protruding through or around the insole and cannot be pulled out, shall be cut close to the surface and shall leave no protruding stumps.



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3.6.22 Heel pad attaching. The flesh side of the heel pad shall be coated with any adhesive specified in 3.2.11 except when upper or gusset leather heel pads are used, the buffed or split side shall be coated with the adhesive. Heel pads shall be pressed firmly and completely into the heel seat. Heal pads shall be fitted in accordance with size schedule.

3.6.23 Laces. The boots shall be mated. A lace shall be inserted through the top eyelet of the outside quarter of each boot, and the two laces shall be tied firmly together.

3.6.24 Repairs of stitching.

3.6.24.1 Type 301 and 401 upper fitting stitches. Repairs of type 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end of stitching.

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area, and continue one stitch beyond the defective area onto the existing stitching.

3.6.24.2 Inseaming and sole stitching. Repairs of inseaming and outsole stitches shall be in accordance with 3.6.24.1 a and b and in addition, thread cuts occuring as a result of trimming operations shall be repaired in the same manner as specified for thread breaks.

3.6.25 Replacement of defective components. During the spreading, cutting, and manufacturing process, components having material defects or damages that are classified as defects in 4.2.2.1.1, 4.2.2.1.2, and 4.2.3.1 shall be removed from production and replaced with nondefective and properly matched components.

3.7 Workmanship. The finished boots shall conform to the quality of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.



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4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

* 4.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

* 4.2.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase documents. In addition, testing shall be performed on components and materials listed in table IX. When data in the "Number of determinations per sample unit" and "Results reported" columns are not specified in table IX, they shall be reported as required by the referenced test methods. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units or the composite sample fails to meet any requirement specified. The sample size shall be as follows:

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |



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TABLE IX. Component tests

| Component and lot size expressed in terms of | Characteristic | Requirement paragraph | Test method | Requirement applicable to: Sample unit | Requirement applicable to: Composite | No. determinations per sample unit | Results reported to nearest | Sample unit |
|---|---|---|---|---|---|---|---|---|
| Insole (Texorist) | Copper-8-quinolinolate | 3.2.1.6 | 2/ | - | - | - | - | - |
| Heel pads (1 pad) | Paranitrophenol | 3.2.1.7 | 6711 1/ | - | X | - | - | 5 heel pads |
| | Material identification | 3.2.1.7 | 2/ | - | - | - | - | |
| | Thickness | 3.2.1.7 | 1011 1/ | X | - | - | - | |
| | Tannage | 3.2.1.7 | 2/ | - | - | - | - | |
| Nails, tacks, and staples | Material identification | 3.2.3.2 and 3.2.3.3 | 2/ | - | - | - | - | - |
| Eyelets 1 pound | Material identification | 3.2.3.4 | 2/ | - | - | - | - | 2 eyelets |
| | Outside diameter of flange and barrel | 3.2.3.4 | Gage | X | - | 2 | 0.001 inch | |
| | Diameter of hole | 3.2.3.4 | Gage | X | - | 2 | 0.001 inch | |
| | Overall length | 3.2.3.4 | Gage | X | - | 2 | 0.001 inch | |
| | Thickness | 3.2.3.4 | 2/ | - | - | - | - | |
| | Finish | 3.2.3.4.1 | 2/ | - | - | - | - | |





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TABLE IX. Component tests (cont'd)

| Component and lot size expressed in terms of | Characteristic | Requirement paragraph | Test method | Requirement applicable to: Sample unit | Requirement applicable to: Composite | No. determinations per sample unit | Results reported to nearest | Sample unit |
|---|---|---|---|---|---|---|---|---|
| Ribbing fabric for insole, 1 yard full width | Material identification | 3.2.4.4.1 | 2/ | - | - | - | - | - |
| | Weight | 3.2.4.4.1 | 5041 2/ 3/ | - | - | - | - | - |
| | Yarns per inch | 3.2.4.4.1 | 5050 2/ 3/ | - | - | - | - | - |
| | Breaking strength | 3.2.4.4.1 | 5100 2/ 3/ | - | - | - | - | - |
| Bottom filler 1 pound | Material identification | 3.2.6 | 2/ | - | - | - | - | - |
| | Solubility of binder in water | 3.2.6.1 and 3.2.6.2 | 2/ | - | - | - | - | - |
| | Softening point of binder | 3.2.6.1 and 3.2.6.2 | Method E 28 4/ | - | 5/ | 2 | Average of 2 determinations to nearest 1°F | Composite 1 pound |
| | Penetration of binder | 3.2.6.1 | Method D 5 4/ | - | 5/ | 3 | Average of 3 determinations to nearest mm | - |





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TABLE IX. Component tests (cont'd)

| Component and lot size expressed in terms of | Characteristic | Requirement paragraph | Test method | Requirement applicable to: Sample unit | Requirement applicable to: Composite | No. determinations per sample unit | Results reported to nearest | Sample unit |
|---|---|---|---|---|---|---|---|---|
| Bottom filler 1 pound (cont'd) | Proportion of cork to binder | 3.2.6.1 and 3.2.6.2 | 2/ | - | - | - | - | |
| Shank board | Material identification | 3.2.7.1 | 2/ | - | - | - | - | |
| | Thickness | 3.2.7.1 | 2/ | - | - | - | - | |
| Steel shank (1 pair) | Material identification | 3.2.7.2 | 2/ | - | - | - | - | |
| | Hardness (Rockwell C scale) | 3.2.7.2 | E 18 4/ | X | - | 1 | Hardness number | 1 steel shank |
| | Coating identification | 3.2.7.2 | 2/ | - | - | - | - | - |
| | Shank width and thickness | 3.2.7.2 | 2/ | - | - | - | - | - |
| Box toes | Material identification | 3.2.8 | 2/ | - | - | - | - | - |
| | Thickness | 3.2.8 | 2/ | - | - | - | - | - |
| | Impregnation compound identification | 3.2.8 | 2/ | - | - | - | - | - |





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TABLE IX. Component tests (cont'd)

| Component and lot size expressed in terms of | Characteristic | Requirement paragraph | Test method | Requirement applicable to: Sample unit | Requirement applicable to: Composite | No. determinations per sample unit | Results reported to nearest | Sample unit |
|---|---|---|---|---|---|---|---|---|
| Pegs, heel seat | Material identification | 3.2.9 | 2/ | - | - | - | - | - |
| | Diameter | 3.2.9 | 2/ | - | - | - | - | - |
| Adhesive | Material identification | 3.2.11 | 2/ | - | - | - | - | - |
| Wax | Material identification | 3.2.12 | 2/ | - | - | - | - | - |

1/ Refers to FED-STD-311.

2/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

3/ Refers to FED-STD-191.

4/ Refers to ASTM standard.

5/ Requirement applicable to lot average.





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4.2.1.1 Examination of components.

* 4.2.1.1.1 Examination of insoles. The leather insoles shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination of KK-I-570. The Texorist insoles shall be examined for defects listed below. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The sample unit shall be one insole. The lot size shall be expressed in units of one insole.

Any hole, cut, tear or gouge.
Any brittle area or evidence of delamination.
Thickness not as specified.
Sole outline not conforming to required pattern.
Sole not cut in specified direction.

4.2.1.1.2 Examination of midsoles. The midsoles shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination of KK-L-165.

4.2.1.1.3 Examination of counters. The counters shall be examined for visual and dimensional requirements conforming to the provisions of the end item examination of MIL-C-41814.

4.2.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or services for which in-process inspection has indicated nonconformance.

* 4.2.2.1 In-process examination. The defects found during in-process examination shall be classified in accordance with 4.2.2.1.1 and 4.2.2.1.2. The inspection level shall be II and the (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability, and minor when affecting serviceability but not seriously.

4.2.2.1.1 Examination of uppers after all fittings. The upper assembly shall be examined for defects listed below which cannot be seen in the end item. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The lot size shall be expressed in units of one upper.



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| Examine | Defect | Classification Major | * | Minor |
|---|---|---|---|---|
| Construction and workmanship (general) | Any component missing or not specified type | X | | |
| | Any component misplaced or not affixed as specified | | * | |
| | Vamp stitched more than 5/32 inch from edge at throat | X | | |
| | Vamp stitched more than 3/32 inch from edge at throat but not over 5/32 inch | | | X |
| | Vamp throat or wings not skived | X | | |
| | Counterpocket-backstay or quarter not skived as specified | | | X |
| | Backseam not rubbed down | | | X |
| Quality of leather | Thickness more than 1/2 ounce less than minimum specified | X | | |
| | Thickness less than specified minimum but not exceeding 1/2 ounce less | | | X |
| | Thickness more than the maximum specified | | | X |
| | Rough fiber on flesh side of quarters | | * | |
| | Off-stretch cut | X | | |
| | Slaughter cut | | * | |

4.2.2.1.2 Examination of boot before bottom filling. The partially fabricated boot shall be examined for defects listed below which cannot be seen in the end item. The sample unit shall be one partially constructed boot at a point after lasting and attachment of shank but before the application of the bottom filler. The lot size shall be expressed in units of one boot.

| Examine | Defect | Classification Major | * | Minor |
|---|---|---|---|---|
| Bottom of boot | Any component missing or not specified type | X | | |
| | Shank not properly positioned or not fitting contour of boot bottom | | * | |
| | Shank wrong size, malformed, or not securely attached | | | X |
| | Any tear in cotton duck in stitch area: | | | |
| | - more than 3/4 inch | X | | |
| | - more than 1/2 inch but not more than 3/4 inch | | | X |



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| Examine | Defect | Classification Major | | Minor |
|---|---|---|---|---|
| Bottom of boot (cont'd) | Any insole tack, staple, or anchor tack not removed | X | | |
| | Excess leather in heel seat interfering with proper fit of shank | | | X |
| | Upper damaged | | * | |
| | Poor heel seat side or toe lasting | | * | |
| | Inseam not properly trimmed | | * | |
| | Less than 3 stitches per inch on inseam | X | | |
| | Less than 3-1/4 but not less than 3 stitches per inch on inseam | | | X |
| | Inseam stitches broken, two or more consecutive skipped inseam stitches, or inseam stitches cut or damaged during trimming operation: | | | |
| | - not repaired | X | | |
| | - not repaired as specified | | | X |
| | Inseam stitches not at bottom of insole rib or not in welt groove | | * | |
| | Broken insole rib | X | | |
| | End of counter not caught in inseam stitching | | | X |
| | Welt not properly beaten out, i.e., not flat | | | X |
| | Welt butts not skived | | | X |
| | Welt butts not tacked within butt area | | | X |
| Upper part of boot | Upper not tightly pulled down to last | | | X |
| | Lace opening less than 7/8 inch or more than 1-1/8 inches | | | X |
| | Quarter not laced as specified | | | X |

4.2.2.2 In-process testing.

4.2.2.2.1 Stuck-on rib insole. When used, the stuck-on rib shall be tested for compliance with the requirements of 3.2.4.4.2. Method 2061 of FED-STD-311 shall be used for shear strength test. Method 2171 of FED-STD-311 shall be used for stitch tear strength. The sample unit shall be two ribbed insoles, one for each test and the sample size shall be five, regardless of the lot quantity. Requirements are applicable to the sample unit for minimum requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all units fail to meet the requirements specified. All test reports shall contain the individual values utilized in expressing the final results.



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4.2.2.2.2 Outsole and midsole assembly. The outsole and midsole assembly shall be tested to determine compliance with the bond strength requirement of 3.6.5. The test method shall be as indicated in 4.3.1. All test reports shall contain the individual values utilized in expressing the final result. The sample unit shall be one cemented midsole to outsole unit. The lot size shall be expressed in terms of one cemented midsole to outsole unit. The requirement is applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet the requirement specified. The sample size shall be as follows:

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

* 4.2.3 End item examination. The end item shall be examined in accordance with 4.2.3.1 and 4.2.3.2.

* 4.2.3.1 Visual examination. The end item shall be examined for the defects listed below. The sample unit shall be one completely fabricated boot and the selection shall be by pairs. The lot size shall be expressed in units of one boot. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. (Note: The AQL does not apply to a protruding point of nail, tack, or staple forward of the heel breast line.) Defects of pairing shall be classified as a single defect. Heel pads removed by the Government during verification examination shall be re-cemented and replaced by the manufacturer's personnel. The vamp and inside and outside quarters of each boot shall be examined for break of leather in accordance with the procedure outlined below:

a. To examine the vamp, the boot shall be held in an upright position with both hands. The top of the boot shall face away from the examiner. Position thumbs on top of vamp approximately halfway between box toe line and blucher noses and 1 inch to 1-1/2 inches apart. Press downward with thumbs so as to form grain surface into a concave surface and observe the break pattern. Any vamp exhibiting a break pattern at any location comparable to a break pattern represented by numbers 5 through 8 on the Satra scale (see 6.5) shall be scored as a defect. The defect shall be scored regardless of the direction in which the break pattern appears.

b. To examine the quarters, the index finger and the second finger held in a "V" position shall be placed against the inside of the quarter, between the fourth and sixth eyelets from top of quarter. The thumb shall be placed on the outside of the quarter over the fifth eyelet. Pressure shall be applied to the quarter with the thumb and fingers so as to impart a break pattern to the leather. The break pattern of the quarter shall be observed and compared with the break patterns of the Satra scale. Any quarter exhibiting a break pattern comparable to a break pattern represented by numbers 5 through 8 on the Satra scale shall be scored as a defect. the following note regarding defects visible at a distance of 3 feet is not applicable to examination of the break pattern.



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NOTE: Regarding conditions affecting appearance, defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more. Defects designated by an asterisk (*) shall be scored as major when seriously affecting serviceability or appearance and minor when affecting serviceability or appearance but not seriously.

| Examine | Defect | Classification Major | | Minor |
|---|---|---|---|---|
| Pairing | Not properly mated, i.e., not right and left of same size | X | | |
| | Variation in color or appearance | | * | |
| | Box toe crooked, long, or short | | * | |
| Cleanness | Any non-removable spot, stain, or foreign matter affecting appearance | | | X |
| Color and finish | Not specified color | X | | |
| | Finish streaky, chipped, or flaky on upper | | | X |
| | Any raw edges not stained to match upper leather | | | X |
| | Sole edges not bright or not finished as specified | | | X |
| | Heel seat or top of welt not finished as specified | | | X |
| Design, type and size | Not as specified | X | | |
| Upper leather | Leather deeply snuffed, i.e., fiber structure damaged | X | | |
| | Any grub or tick damage, scratch, brand, or boney leather | | * | |
| | Fat wrinkles, or veins | | * | |
| | Stretchy vamp | X | | |
| | Leather embossed or printed | X | | |
| Break pattern of vamps and quarters | Break pattern represented by numbers 5 through 8 on Satra break scale | X | | |
| | NOTE: Leather exhibiting a break pattern comparable to the numbers indicated above is considered to be loose, pipey, or flanky. The break shall be determined in accordance with 4.2.3.1. | | | |



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| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Construction and workmanship (general) | Any cut, tear, hole, repair, or factory damage | * | |
| | Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise classified herein) | * | |
| | Wrinkled or bunched area at the back seam | * | |
| | Loose leather fibers exposed along cut edge of quarter vamp line or counter-pocket-backstay | | X |
| Seams and stitching (upper) | Open seam not repaired | X | |
| | Repair of open seam not as specified | | X |
| NOTE: | A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When any of the above conditions occurs on the inside of the boot, it shall be scored as an asterisk (*) defect | | |
| | Loose tension resulting in a loosely secured seam | * | |
| | Tight tension resulting in puckering or cutting of leather | * | |
| | Wrong stitch or seam type | X | |
| | Any row of stitching with less than 8 stitches per inch but not less than 6 stitches per inch | | X |
| | Any row of stitching with less than 6 stitches per inch | X | |
| | More than the specified maximum number of stitches | * | |
| NOTE: | A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners. | | |
| | Gage of stitching not as specified or irregular | * | |
| | Stitching omitted where required | * | |
| | Needle holes or needle chew | * | |
| | Thread ends not trimmed throughout boot | | X |
| | Sand hole, i.e., tongue not properly caught in at throat | | X |



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| Examine | Defect | Classification Major | | Minor |
|---|---|---|---|---|
| Outsole stitching | Less than 5 stitches per inch or more than 9 stitches per inch | X | | |
| | Less than 5-1/2 stitches per inch but not less than 5 stitches per inch, or more than 7-1/2 stitches per inch but not exceeding 9 stitches per inch | | | X |
| | Broken stitch or two or more consecutive skipped stitches: | | | |
| | - not repaired | X | | |
| | - not repaired as specified | | | X |
| | Stitches short at heel breast | | * | |
| | Stitch locks laying on surface | | * | |
| | Outsole or welt damaged by stitches | X | | |
| Outsole plaque (tread design) | Off center by more than 1/4 inch | X | | |
| Counters | Rolled or curled counter | | * | |
| | Soft counter | | | X |
| Edge making | Edge trimmed into sole stitching | | * | |
| | Edge not trimmed half round | | | X |
| | Edge trimming is irregular affecting appearance | | | X |
| | Sole extension less than specified minimum | | * | |
| | Edge setting not as specified | | * | |
| | Checked sole, i.e., separation between outsole and midsole | | * | |
| Heel finishing and attaching | Heel not finished square, i.e., has pronounced flare or taper | | | X |
| | Heel not finished smooth affecting appearance | | | X |
| | Open heel seat | | * | |
| | Checked heel, i.e., separation at heel and outsole | | | X |
| | Crooked or wrong size heel | | | X |
| Sole reinforcing nails | One or more reinforcing nails missing | | | X |
| | Nails not driven slightly below surface of rubber sole | | | X |
| | Incorrect nailing pattern | | | X |
| | Nails not properly clinched on insole | | | X |



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| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Eyelets | Number of eyelets not as specified but each row has the same number | | X |
| | Not the same number of eyelets in each row | X | |
| | Eyelets not properly spaced within the row or misalignment between the rows to an extent interfering with proper lacing | * | |
| | Edge of any eyelet less than 1/16 inch from the edge of the quarter | X | |
| | Edge of any eyelet less than 1/8 inch but not less than 1/16 inch or more than 5/16 inch from the edge of the quarter | | X |
| | Any eyelet not securely clinched | | X |
| Inseaming | Grinning seam, i.e., thread exposed | X | |
| | Strained seam, i.e., needle holes visible but thread not exposed | | X |
| | Dropped welting, skipped stitches, or cut inseam | X | |
| Lining | Vamp lining having excessive fullness; torn, loose, or wrinkled | * | |
| Insole | Short or long | * | |
| | Any protruding point of lasting staple, tack, or nail forward of heel breast line (see footnote 1/) | | |
| | Any protruding point of tack, fastener peg, nail, or point of lasting staple in heel seat area | X | |
| | No evidence of heel seat fastening | | X |
| | More than three heel seat fastener pegs or nails run completely off the insole | X | |
| | More than three heel seat fastener pegs or nails not properly positioned to edge of insole | | X |
| Heel pad | Not firmly and adequately adhered to heel seat | | X |
| Marking | Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size | | X |

1/ Any protruding point of lasting staple, tack, or nail forward of the heel breast line, found in the sample, shall cause rejection of the lot represented.



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* 4.2.3.2 Heel nailing examination. The heel nailing shall be exmined for the defects listed below. The sample unit shall be one boot. The lot size shall be expressed in units of one boot. The inspection level shall be I and the AQL, expressed in terms or defects per hundred units, shall be 6.5.

| Examine | Defect |
|---|---|
| Heel nailing | More than three nails unclinched or runoff insole (does not grip insole)<br>Either or both breast nails unclinched or runoff insole<br>More than two of five back nails unclinched or runoff insole<br>Any heel nail missing |
| NOTE: | Evidence of brass or steel on the insole shall indicate clinching. |

4.2.4 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully packed. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

| Examine | Defect |
|---|---|
| Marking (exterior) | Omitted; incorrect; illegible; of improper size, location, sequence, or method of application |
| Materials | Any component missing, damaged, or not as specified |
| Workmanship | Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling<br>Bulged or distorted container |
| Content | Number of pairs of boots per container is more or less than required |

4.3 Methods of inspection.

4.3.1 Bond strength of outsole to midsole.



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4.3.1.1 Specimen. The specimen shall be a combined midsole-outsole unit which has aged at least 2 days. The outsole shall be separated from the midsole for a distance of approximately 2-1/2 inches from the toe end of the specimen.

4.3.1.2 Apparatus. A power driven portable gage tong adhesion machine or an approved portable testing device of equal performance shall be used. The rate of travel of the power actuated grip shall be 2 inches per minute. The machine shall be operated without any device for maintaining maximum load indication.

4.3.1.3 Procedure. The separate toe ends of the specimen shall be clamped in the jaws of the machine, with the jaws 2 inches apart. The specimen shall extend outward at right angles to the direction of the application load. The machine shall be started and the outsole and midsole shall be pulled apart for a distance of 1 inch. At that instant, the load indicated on the machine shall be read and recorded. The load shall be divided by the width of the specimen at the corresponding line of separation.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each pair of properly mated boots shall be tied together by the ends of laces which shall be inserted through the top eyelet of the outside quarter of each boot.

5.1.2 Commercial. Boots shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table X (B) and packed in a fiberboard shipping container conforming to style FOL-L, grade V2s of PPP-B-636. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636, except that the joint may be secured in accordance with commercial practice. The approximate inside dimensions of the shipping container shall be as specified in table X (A). Each container shall have the contents completely covered on the top and bottom with a sheet of 30-pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same container.



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5.2.2 Level B packing. Twelve pairs of boots of one size and width only, preserved as specified in 5.1, shall be arranged as shown in table X (B) and packed in a fiberboard shipping container conforming to FOL-L, type CF, class domestic, variety SW, grade 275 of PPP-B-636. The manufacturer's joint shall be metal stitched. The inside of each fiberboard container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636, except that the joint may be secured in accordance with commercial practice. The approximate inside dimensions of the shipping container shall be as specified in table X (A). Each container shall have the contents completely covered on the top and bottom with a sheet of 30 pound minimum basis weight kraft paper conforming to type I, grade B of UU-P-268. Each container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.2.4. Toward the end of the contract or when there are less than the required amount per container of the same size and width, mixed sizes and widths may be packed in the same container.

5.2.2.1 Weather resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in PPP-B-636, except that the inspection shall be in accordance with 4.2.4.

TABLE X. Shipping container dimensions and packing arrangement

(A) Approximate inside dimensions of shipping containers 1/

| Range of boot sizes | Length inches | Width inches | Depth inches |
|---|---|---|---|
| 5XN through 8-1/2XW | 30-1/2 | 12-1/2 | 13 |
| 9XN through 12XW | 30-1/2 | 13-1/4 | 13-3/4 |

1/ Approximate dimensions are furnished as a guide only. For boot sizes below 5XN and above 12XW, the contractor shall determine the size container to use. The dimensions of the container shall be such that it will result in a snug fit of the contents to be packed therein. The use of blocking materials to accommodate small size boots in an oversize container will not be permitted.

(B) Packing arrangements. Arrangement of boots within the shipping container shall be in four layers of three pairs per layer on their sides with the first layer started by placing the left boot in the left-hand end of the container with the heel toward the front and the sole facing left, and the right boot laid over the quarter of the left boot in the same manner.

Two more pairs shall be placed in a like manner to complete the first layer. Care shall be taken that the first layer is properly spaced in order to facilitate the packing of the other three layers. The second layer shall start at the left by placing the right boot with the heel toward the rear, the top facing left, and the sole face to face with the heel of the second boot in the first layer, and so on to complete the layer. The third layer shall be placed exactly as the first, and the fourth layer shall be the same as the second. The last boot placed in the first and third layers, and the first boot placed in the second layer shall have the tops turned upward. The first boot placed in the top layer shall have the top folded under. (See figure 3)



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5.2.3 Commercial packing. The boots, preserved as specified in 5.1, shall be packed in accordance with MIL-STD-1188.

5.3 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or MIL-STD-1188, as applicable.

5.3.1 Width marking. Following the item description and size designation, in the same lettering, the width shall be spelled out; i.e., EXTRA NARROW, NARROW, REGULAR, WIDE, OR EXTRA WIDE.

5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes and widths shall have securely attached to the side and end directly under the printing or stenciling, a white paper label 5 by 4 inches with the words MIXED NSNs and WIDTHS clearly stamped or printed thereon and under these words shall be legibly stamped or printed the correct quantity of pairs, NSNs, and widths contained therein.

6. NOTES

6.1 Intended use. The boots are for use by Department of Defense military personnel requiring orthopedic correction of their footwear.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this document.
(b) Sizes and widths required (see 1.2).
(c) Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
(d) When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 Recycled materials. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.2).

6.5 Satra scale. The Satra scale may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England, or may be obtained from Satra House's North American agent: Bata Engineering, Batawa, Ontario Canada.

6.6 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Army - GL
Navy - NU
Air Force - 99

Review activities:

Army - MD
Air Force - 45
DLA - CT

User activities:

Navy - CG, MC

Preparing activity:

Army - GL

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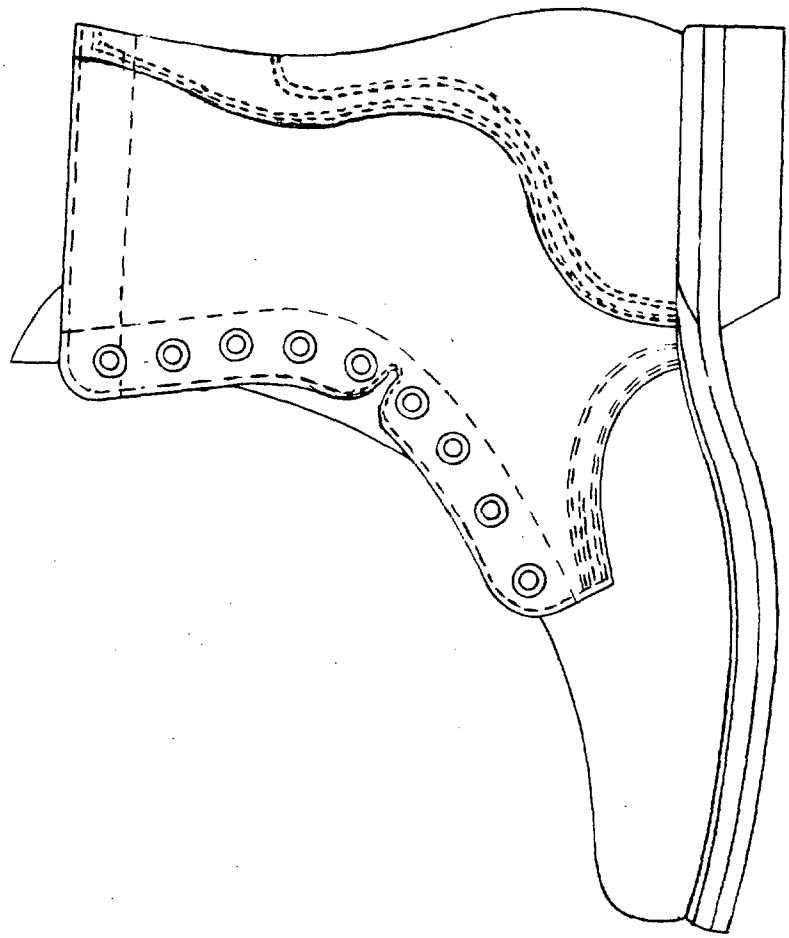


FIG 1. BOOTS, COMBAT, LEATHER, BLACK 21-608F



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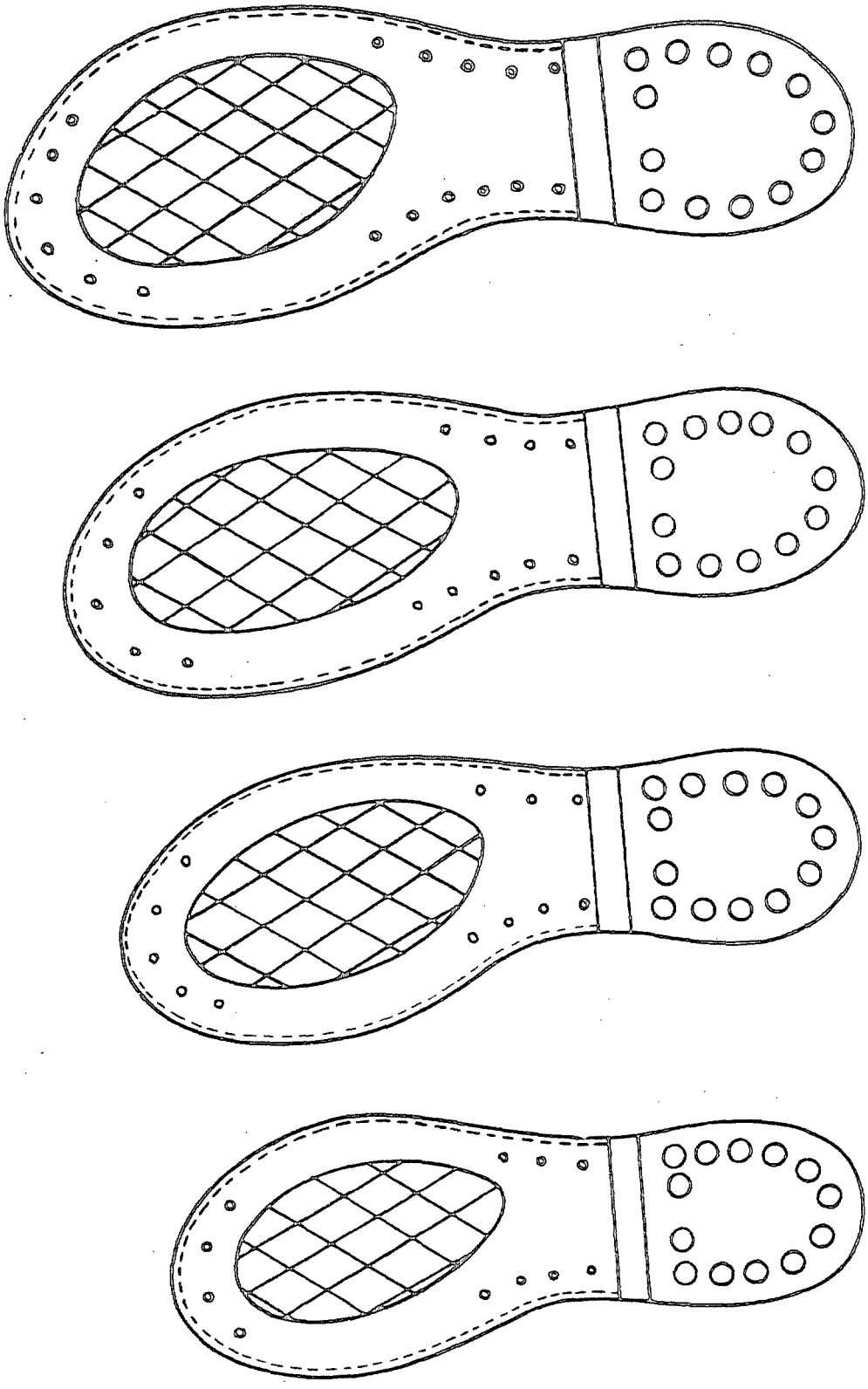



FIG. 2 BOOT, COMBAT, LEATHER, BLACK; 2-1-41E
SOLE REINFORCING, (NAILING PATTERN)



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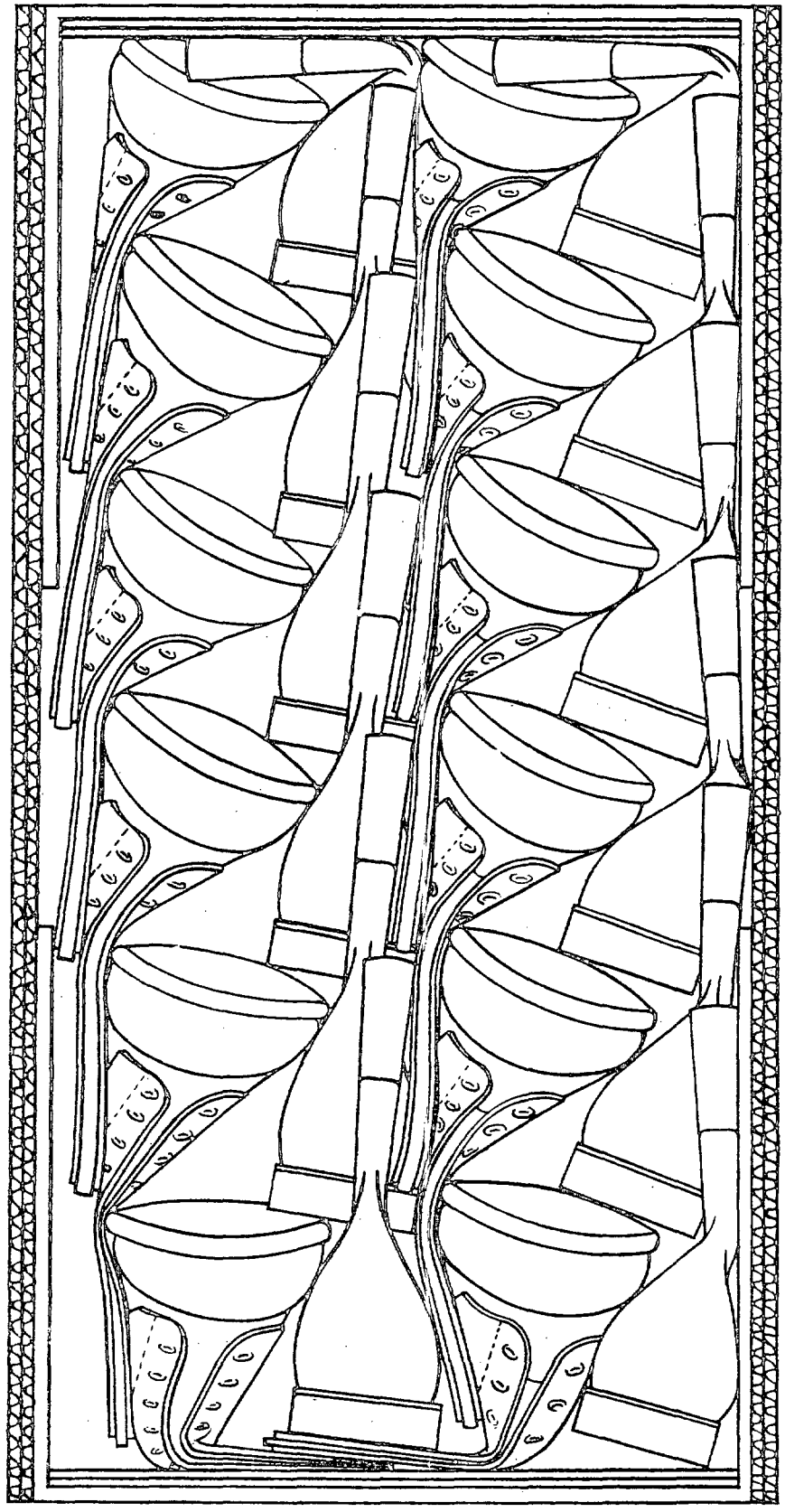


FIG. 3 BOOT ARRANGEMENT WITHIN SHIPPING CONTAINER (SIDE REMOVED)



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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-B-41818G Boots, Combat, Leather, Black, (MIL-5)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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C. REASON FOR RECOMMENDED CHANGE(S)

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